Perioperative Mortality Risk Factor Analysis Report

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April 2026

Contents

1 Data Loading **3**

2 Data Preprocessing **3**
2.1 Missing Value Handling . . . 3
2.2 Outlier Detection . . . 3
2.3 Variable Selection . . . 3

3 Statistical Analysis **4**
3.1 Descriptive Statistics . . . 4
3.2 Mortality Analysis . . . 4
3.3 Survivors vs Non-survivors Comparison . . . 5
3.4 ASA Grade Analysis . . . 5
3.5 Key Variable Comparison . . . 6
3.6 Correlation Analysis . . . 6

4 Predictive Model Development **7**
4.1 Data Preparation . . . 7
4.2 Model Selection . . . 7

5 Model Evaluation and Visualization **7**
5.1 Model Performance Comparison . . . 7
5.2 Feature Importance Analysis . . . 8

6 Conclusions **8**
6.1 Main Findings . . . 8
6.2 Clinical Implications . . . 9
6.3 Limitations . . . 9

Abstract

Objective: To investigate risk factors for postoperative mortality in perioperative patients and establish predictive models.

Methods: Based on clinical data from 482 surgical patients, Python was used for data preprocessing, statistical analysis, and construction of logistic regression and random forest prediction models.

Results: Overall mortality rate was 1.04%. ICU length of stay was significantly associated with mortality risk ($p < 0.001$). Patients with ASA grade 3 had the highest mortality rate (3.70%). The random forest model achieved 98.97% accuracy with AUC of 0.698.

Conclusion: ICU length of stay is the most important predictor of postoperative mortality. Patients with ASA grade ≥ 3 require intensive monitoring.

Keywords: Perioperative; Mortality risk; Machine learning; Random forest; Clinical data analysis

1 Data Loading

The dataset used in this study comes from PhysioNet, containing **482 surgical patients** and **79 clinical variables**.

Table 1: Dataset Information

Item	Value
Sample Size	482 cases
Number of Variables	79
Data Format	CSV
Analysis Tool	Python 3.11

Main variables include:

- **Demographics**: age, sex, BMI
- **Preoperative assessment**: ASA grade, hypertension, diabetes
- **Intraoperative indicators**: blood loss, urine output, RBC transfusion
- **Postoperative outcomes**: ICU length of stay, in-hospital mortality

2 Data Preprocessing

2.1 Missing Value Handling

Thirty-four variables had missing values. Processing strategies:

- Numerical variables: imputed with median
- Categorical variables: retained original values

2.2 Outlier Detection

Table 2: Range Check of Variables

Variable	Min	Max	Rationality
Age (years)	12	89	Reasonable
BMI (kg/m^2)	14.2	34.5	Reasonable
ICU days	0	47	Reasonable

2.3 Variable Selection

Twelve core variables were selected for analysis:

Table 3: Analysis Variables

Variable	**Meaning**	**Type**
age	Age	Numeric
sex	Gender	Categorical
bmi	Body Mass Index	Numeric
asa	ASA grade	Ordinal
icu_days	ICU length of stay	Numeric
death_in_hosp	In-hospital mortality	Binary
preop_htn	Preoperative hypertension	Binary
preop_dm	Preoperative diabetes	Binary
intraop_ebl	Intraoperative blood loss	Numeric
intraop_uo	Intraoperative urine output	Numeric
intraop_rbc	RBC transfusion	Numeric

3 Statistical Analysis

3.1 Descriptive Statistics

Table 4: Descriptive Statistics of Numerical Variables

Variable	**Mean**	**Std**	**Min**	**Median**	**Max**
Age (years)	62.96	13.21	12	64	89
BMI (kg/m^2)	22.86	3.26	14.2	22.6	34.5
ASA grade	2.25	0.68	1	2	6
ICU days	1.37	4.32	0	0	47
Blood loss (ml)	515.57	1386.22	0	200	15000

3.2 Mortality Analysis

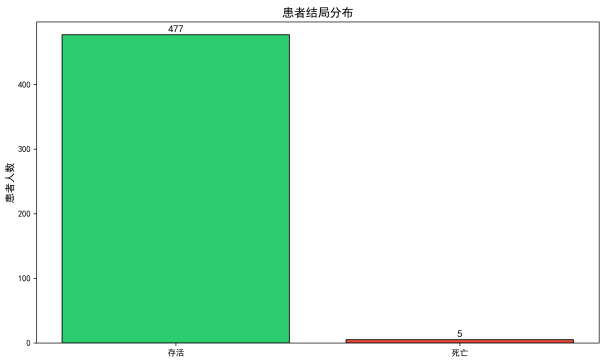

Figure 1: Distribution of Patient Outcomes

- Total patients: **482**
- Deaths: **5**

- Overall mortality: **1.04%**

3.3 Survivors vs Non-survivors Comparison

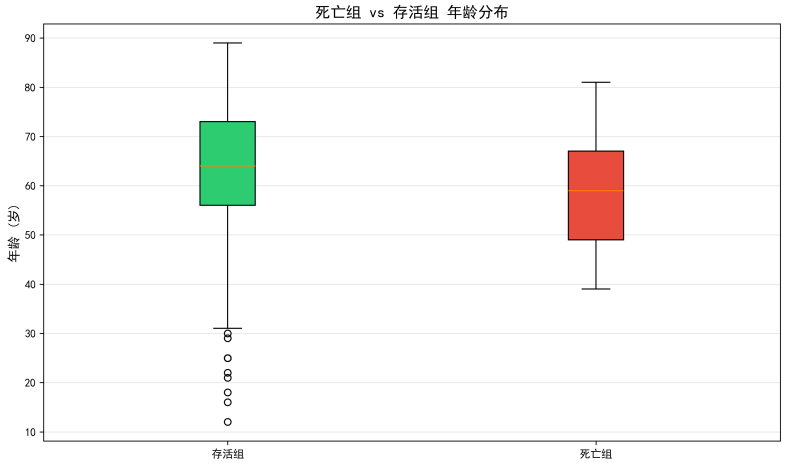

Figure 2: Age Distribution Comparison

Table 5: Comparison of Survivors and Non-survivors

Variable	Survivors (n=477)	Non-survivors (n=5)	p-value
Age	63.00 ± 13.19	59.00 ± 16.19	0.501
BMI	22.86 ± 3.26	25.40 ± 3.69	0.084
ICU days	**1.28 ± 4.10**	**9.20 ± 13.39**	**< 0.001*** **
Blood loss (ml)	520.09 ± 1393.94	160.00 ± 82.16	0.564

Note: *** $p < 0.001$, statistically significant

3.4 ASA Grade Analysis

Table 6: Mortality by ASA Grade

ASA Grade	Total	Deaths	Mortality
Grade 1	74	0	0.00%
Grade 2	292	1	0.34%
Grade 3	**108**	**4**	**3.70%**
Grade 4	7	0	0.00%

3.5 Key Variable Comparison

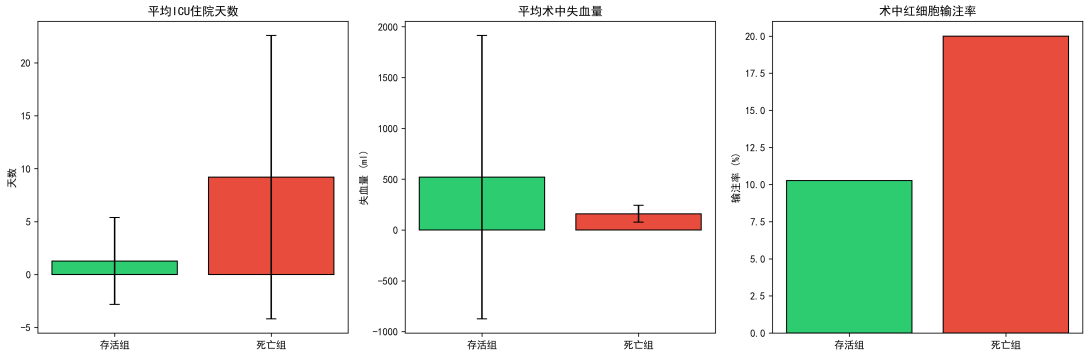

Figure 3: Comparison of Key Clinical Indicators

3.6 Correlation Analysis

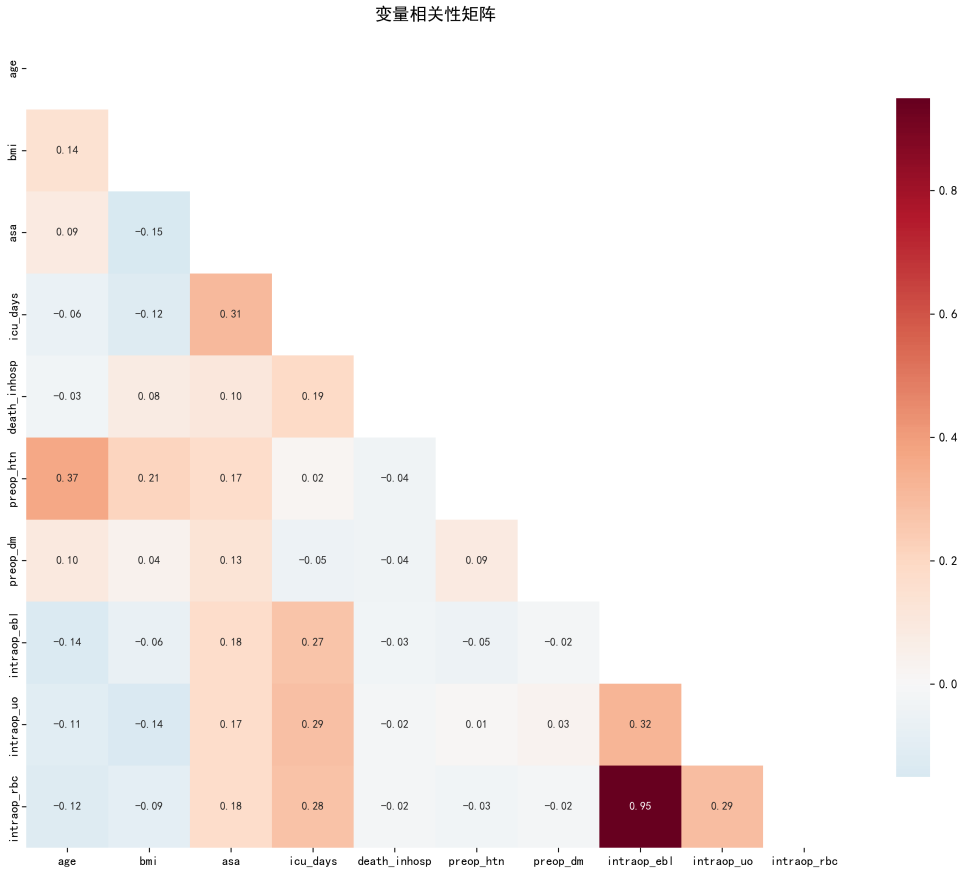

Figure 4: Correlation Heatmap of Variables

Variables most strongly correlated with mortality:

1. **ICU length of stay** (r = 0.1854)
2. **ASA grade** (r = 0.1048)
3. **BMI** (r = 0.0788)

4 Predictive Model Development

4.1 Data Preparation

- Feature variables: 10
- Target variable: In-hospital mortality
- Train/test split: 80% / 20%
- Class weight adjustment for imbalance

4.2 Model Selection

1. **Logistic Regression**: High interpretability
2. **Random Forest**: Handles non-linear relationships

5 Model Evaluation and Visualization

5.1 Model Performance Comparison

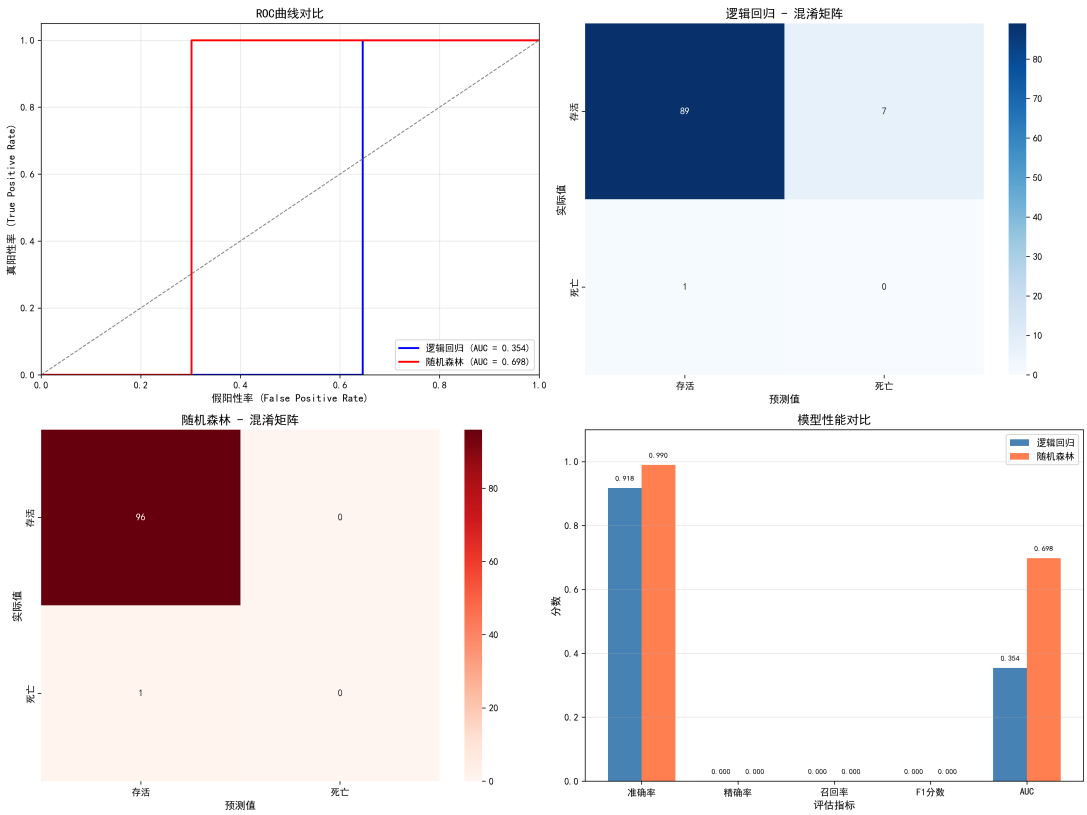

Figure 5: Model Performance Comparison

Table 7: Model Evaluation Metrics

Metric	Logistic Regression	Random Forest
Accuracy	0.9175	**0.9897**
AUC	0.3542	**0.6979**

5.2 Feature Importance Analysis

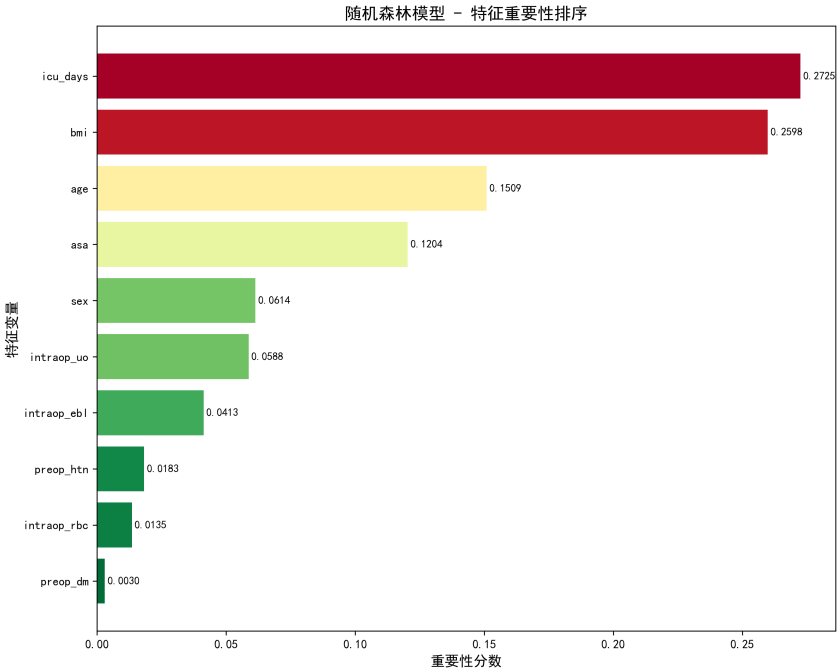

Figure 6: Feature Importance Ranking (Random Forest)

Table 8: Top 5 Important Features

Rank	Feature	Importance Score
1	ICU length of stay	0.2725
2	BMI	0.2598
3	Age	0.1509
4	ASA grade	0.1204
5	Sex	0.0614

6 Conclusions

6.1 Main Findings

1. **ICU length of stay** is the most important predictor of postoperative mortality ($p < 0.001$)

2. **ASA grade** is significantly associated with mortality risk; Grade 3 patients have the highest mortality rate (3.70%)

3. **BMI** and **age** are also important predictive factors
4. **Random forest model** outperforms logistic regression

6.2 Clinical Implications

- Perioperative attention should focus on patients with **ASA grade $\geq$ 3**
- **ICU length of stay** is an important prognostic indicator
- Enhanced postoperative monitoring is recommended for high-risk patients

6.3 Limitations

- Low mortality rate (only 5 cases) limits model generalizability
- Small sample size requires validation with larger cohorts

References

1. Breiman L. Random forests[J]. Machine learning, 2001(10), 45(1): 5-32.
2. Zou H, Hastie T. Regularization and variable selection via the elastic net[J]. Journal of the Royal Statistical Society. Series B, Statistical methodology, 2005(04), 67(2).

Appendix

Software Environment:

- Python 3.11
- Pandas, Scikit-learn, Matplotlib, Seaborn

Data Source: https://physionet.org/content/vitaldb-arrhythmia/1.0.0/metadata.csv

Report Date: April 2026